

Epic

Build and automate CI workflow for Retail Order Management Platform

Trello Board Lists

- Backlog
 - Features
 - User Stories
 - Tasks
 - In Progress
 - Completed
-

Features

Feature 1 — Develop Flask APIs

Description

Develop Flask APIs for:

- Health Check
 - Order Status
 - Inventory Status
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Feature 2 — Automated API Testing

Description

Implement automated testing using pytest.

Feature 3 — GitHub Repository Management

Description

Manage project source code using GitHub.

Feature 4 — GitHub Actions CI Pipeline

Description

Create CI pipeline for automated testing and validation.

Feature 5 — Build Verification and Monitoring

Description

Verify build success and monitor pipeline execution.

User Stories

User Story 1 — Health API Monitoring

Description

As an admin,
I want a health API,
So that I can monitor service availability.

User Story 2 — Order Status Tracking

Description

As a customer,
I want order status updates,
So that I can track order progress.

User Story 3 — Inventory Availability

Description

As warehouse staff,
I want inventory details,
So that stock can be managed efficiently.

User Story 4 — Automated Testing

Description

As a QA engineer,
I want automated testing,
So that APIs are validated automatically.

User Story 5 — CI Automation

Description

As a DevOps engineer,
I want CI automation,
So that tests run automatically on every push.

Tasks

Task 1 — Flask Application Development

Checklist

- Create Flask project structure
- Create app.py
- Develop health endpoint
- Develop order-status endpoint
- Develop inventory endpoint
- Create requirements.txt
- Run Flask application locally

Task 2 — Automated Testing

Checklist

- Install pytest
 - Create test_app.py
 - Write API test cases
 - Run pytest locally
 - Verify all tests pass
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Task 3 — GitHub Repository Setup

Checklist

- Create GitHub repository
 - Initialize git
 - Add project files
 - Commit source code
 - Push code to GitHub
 - Create README.md
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Task 4 — GitHub Actions CI Pipeline

Checklist

- Create `.github/workflows` folder
 - Create `python-app.yml`
 - Configure Python setup
 - Install dependencies automatically
 - Run pytest in workflow
 - Verify CI execution
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Task 5 — Build Verification

Checklist

- Verify pipeline success

- Check test results
- Capture screenshots
- Validate APIs
- Move completed tasks